

Near-Field Beam Training With Sparse DFT Codebook

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Abstract—Extremely large-scale arrays (XL-arrays) have emerged as one promising technology to improve the spectral efficiency and spatial resolution in future sixth generation (6G) wireless systems. The drastic increase in the number of antennas renders the communication users more likely to be located in the near-field region, which requires a more accurate spherical (instead of planar) wavefront propagation modeling. However, this also inevitably incurs unaffordable beam training overhead when performing a two-dimensional (2D) beam-search in both the angular and range domains. To address this issue, we first introduce in this paper a new *sparse* discrete Fourier transform (DFT) codebook, which exhibits the angular periodicity in the received beam pattern at the user. This thus motivates us to propose a three-phase beam training scheme. Specifically, in the first phase, we utilize the sparse DFT codebook for beam sweeping in an angular subspace and estimate candidate user angles according to the received beam pattern. Then, a central subarray is activated to scan specific candidate angles for resolving the issue of angular ambiguity for identifying the user angle. In the third phase, the polar-domain codebook is applied in the estimated angle to search the best effective user range. Finally, numerical results show that our proposed beam training scheme enabled by the sparse DFT codebook achieves 98.67% beam training overhead reduction as compared to the exhaustive-search scheme, yet without compromising rate performance in the high signal-to-ratio (SNR) regime.

Index Terms—Extremely large-scale array, near-field communications, beam training, DFT codebook, sparse array.

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I. INTRODUCTION

EXTREMELY large-scale arrays/surfaces (XL-arrays/surfaces) have been envisioned as one of the key ingredients to drive the evolution of sixth generation (6G) wireless systems [1], [2]. Specifically, XL-arrays/surfaces with a significant number of antennas can be deployed at the base station (BS) to achieve ultra-high spectral efficiency and spatial resolution, hence accommodating the escalating demands for new applications such as spanning metaverse and digital twin [3], [4]. The drastic increase in the number of antennas in high-frequency bands represents a qualitative paradigm shift in the electromagnetic (EM) propagation modeling, giving rise to the new *near-field communications* [5].

Particularly, different from the far-field EM propagation which is simply approximated by planar waves, the near-field channel modeling necessitates the use of more accurate spherical waves [6], [7]. As such, near-field communications possess several unique properties in contrast to far-field communications. First, the spherical wavefront characteristic opens up the possibility of near-field *beamfocusing*, for which the beam energy can be concentrated at a specific location/region rather than a spatial angle typically for far-field beamforming [8]. The beam-focusing capability of near-field communications enables XL-array to flexibly form highly directional beams in both the angle and range domain, and hence can be leveraged in various applications to improve the system performance, such as mitigating the inter-user interference, improving the accuracy of sensing and localization [9], [10], and enhancing the charging efficiency of wireless power transfer [11]. Second, the rank of line-of-sight (LoS) channels for near-field multiple-input multiple-output (MIMO) communication systems can be larger than one, hence enhancing the spatial multiplexing gains [12]. However, due to the mismatch between the frequency-dependent channel and frequency-independent phase shifter (PS) based beamforming, a significant near-field beam-split effect will be introduced in the wide-band communication systems with huge bandwidth, resulting in the achievable rate performance degradation [13]. This phenomenon can be overcome by true-time-delay (TTD) beamforming architectures due to frequency-dependent phase shifts provided by TTD circuits [13].

In this paper, we propose an efficient near-field beam training scheme with an sparse discrete Fourier transform (DFT) codebook by using the sparse antenna activation method, which significantly reduces the beam training overhead.

A. Related Works

1) *Near-Field Wireless Systems*: Near-field communications bring new opportunities and challenges, which has motivated upsurging research interest. For example, in [14], the authors studied the beamfocusing design based on the fully-digital architectures, hybrid PS-based precoders, and dynamic metasurface antenna architecture for XL-arrays. It is shown that near-field beamfocusing provides new degree of freedom (DoF) to mitigate the interference among users, even when they are located at the same spatial angle. In addition, a new concept of location division multiple access (LDMA) was proposed in [15], which exploits orthogonality of the near-field beamfocusing vectors in the range domain to serve different users at the same angle.

Beyond SWs, XL-array introduces spatial non-stationarity in the near-field communications, where antennas in the XL-array with different spatial locations observe different channel features such as multi-path characteristics and channel gains. The authors in [16] investigated a low-complexity channel model for non-stationary near-field channels, while the authors in [17] developed a framework for analyzing and designing XL-array systems with spatial non-stationarity, for which the performance of the proposed framework approaches that of the conventional full-antenna array based designs albeit with lower complexity. In addition, a holographic metasurface antennas (HMAs) based multi-user system was investigated in [18], where the digital transmit precoder and the analog HMA weighting matrix were jointly optimized to minimize the transmit power. The authors in [19] considered a simultaneous wireless information and power transfer (SWIPT) system, where energy harvesting (EH) and information decoding receivers are located in the near- and far-field regions of the XL-array, respectively. The beam scheduling and power allocation were jointly optimized to maximize the weighted-sum power harvested at EH receivers [19]. Moreover, a directional modulation system was developed for near-field physical layer security systems [20]. Specifically, a fully analog precoding algorithm along with artificial noise and power allocation was proposed to realize secure transmission in both the angular and range domains. Near-field sensing with XL-array was studied in [21], where the closed-form expressions of the Cramér-Rao Bounds for both the angle and range estimations are derived. For the near-field channel estimation, an efficient damped Newtonized orthogonal matching pursuit algorithm associated with the planar and spherical wave models was proposed in [22], for which the angle and range parameter were decoupled and estimated via an central subarray and discrete active elements, respectively. Compared with the least square based channel estimation method, the proposed method achieves comparable accuracy with low pilot overhead and computational complexity. In addition, by leveraging the spatial non-stationarity, the authors in [23] proposed to convert the non-stationary channel to a series of spatial stationary channels, which were estimated by compressed sensing (CS) based techniques and significantly reduce the pilots.

2) *Near-Field Beam Training*: In high-frequency bands, direct channel estimation methods may not be very effective

due to severe path-loss and signal misalignment. As such, beam training is efficient in establishing initial links with high signal-to-noise ratio (SNR) for data transmission and channel state information (CSI) acquisition [24]. However, due to the spherical wave propagation, near-field beam training is more challenging compared with its far-field counterpart as it requires a joint beam search over both the angular and range domain. Particularly, conventional far-field beam training will suffer from significant performance loss in the near-field region due to the so-called *energy-spread effect*, for which the energy of a far-field beamformer is no longer steered towards one angle, but spread in multiple angles. Hence, the conventional far-field beam training method cannot be directly applied to the near-field beam training. To address this issue, the authors in [25] proposed a new codebook design in the polar domain, for which the angular domain is uniformly sampled whereas the range domain is non-uniformly sampled. One can simply invoke this codebook for the exhaustive-search based near-field beam training. However, the overhead of this scheme is the product of the number of antennas and range samples, which is prohibitively high for the implementation of XL-arrays. To reduce the overhead, the authors in [26] proposed a novel two-phase near-field beam training method. This method leveraged a key observation that the true user angle approximately lies in the middle of an angular support with high received SNRs. Based on this observation, one can first estimate the user angle by using far-field DFT beams, and then estimate the user range with the polar-domain codebook in [25]. Besides, deep learning techniques have also been exploited in [27] and [28] to reduce the near-field beam training overhead, where deep neural networks (DNN) are trained based on conventional far-field codebooks and near-field codebook, respectively. However, the training overhead of these methods scales linearly with the number of antennas, which is still unaffordable for communication systems. This issue motivates the design of efficient hierarchical beam training schemes for near-field communications to reduce the training overhead to the logarithmic order, e.g., [29], [30], [31]. However, hierarchical beam training schemes suffer from several inherent drawbacks. First, hierarchical methods usually require frequent feedback and neglect underlying transmission delays. Moreover, there exists the error propagation issue due to the progressive beam search [32].

B. Motivations and Contributions

The existing works on near-field beam training inevitably incur unacceptable overhead, while the off-grid channel estimation methods face highly computational complexity such as high dimensional matrix inversion. Moreover, the beam training overhead of these existing works scales linearly with the number of antennas apart from hierarchical beam training schemes, which suffers from several inherent drawbacks such as error propagation and user feed-back overhead [32]. Motivated by the above, this paper explores a new sparse DFT codebook and a three-phase beam training scheme to reduce the near-field beam training overhead, which scales with the square root of the number of antennas. The main contributions are summarized as follows.

TABLE I
LIST OF MAIN SYMBOLS AND THEIR PHYSICAL MEANINGS

N	Number of BS antennas	U	Antenna activation interval
D	Array aperture size	M	Number of antennas of the activated subarray
λ	Carrier wavelength	z	AWGN
d_0	Antenna spacing	Q	Number of antennas of the activated SLA
$\mathbf{h}_{\text{near}}^H$	Near-field channel	κ	Rician factor
L	Number of channel paths	P_{tot}	Transmit power
θ_0	BS center-user spatial angle	$\mathcal{W}_{\text{DFT}}$	Sparse DFT codebook
r_0	BS center-user range	β	Channel gain
$\mathbf{b}(r_0, \theta_0)$	Near-field channel steering vector	Δ	Difference of spatial angles
$\mathbf{w}$	Beamforming vector	$f(r_0, \theta_0; \theta)$	Received beam pattern
$\mathcal{V}_{\text{Sub}}$	DFT codebook for the activated subarray	$\bar{\mathcal{X}}_{\text{Pol}}$	Polar-domain codebook

- First, we propose a novel sparse DFT codebook by sparsely activating the XL-array antennas and constructing the reduced DFT codebook with the equivalent sparse linear array (SLA). Specifically, the sparse DFT codebook consists of sparse far-field channel response vectors, which is sampled from far-field channel response vectors. Then, we characterize the received beam pattern at the near-field user when the sparse DFT codebook is used for beam sweeping. Interestingly, it is shown that the received beam pattern exhibits periodicity in the angular domain, while there still exists the energy-spread effect. Then, we show that the user angle information is contained in a period of the received beam pattern at the user and can be estimated via a defined *angular support*.
- Second, we propose a novel three-phase beam training scheme based on the sparse DFT codebook. Specifically, in the first phase, we utilize a small number of the sparse DFT codewords to sweep an angular subspace and estimate one candidate user angle according to the middle of the defined angular support. Then, in the second phase, we activate a central subarray to resolve the angular ambiguity by virtue of the periodicity of the received beam pattern at the user. Subsequently, the polar-domain codebook is utilized to search the best user range in the estimated user angle.
- Finally, extensive numerical results are presented to demonstrate the effectiveness of our proposed beam training scheme enabled by the sparse DFT codebook. It is shown that the proposed three-phase beam training scheme can achieve nearly the same performance with the exhaustive-search beam training method in the high-SNR regime, while reducing more than 98% of the training overhead. In the low-SNR regime, the proposed scheme suffers from slight performance loss, while the effective rate still significantly exceeds all benchmark schemes due to lower beam training overhead.

C. Organization and Notations

The remainder of this paper is organized as follows. System model is presented in Sections II. In Section III, we introduce several benchmarks. Section IV provides comprehensive

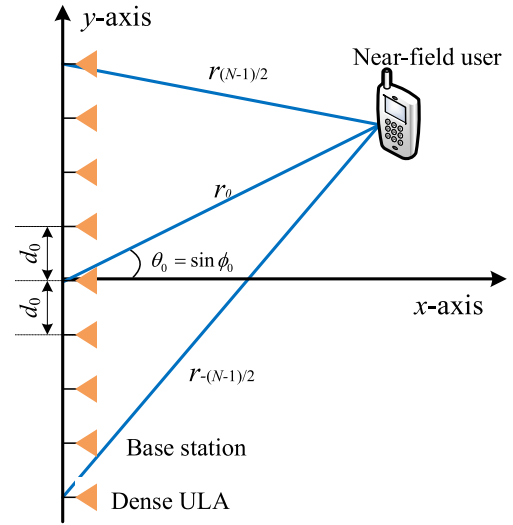


Fig. 1. A narrow-band Near-field XL-array communication system.

analysis of the received beam pattern at the user with the sparse DFT codebook. Then, in Section V, the proposed three-phase beam training scheme is elaborated. Finally, numerical results are provided in section VI to demonstrate the effectiveness of the proposed beam training scheme followed by the conclusions made in Section VII.

Notations: Vectors and matrices are respectively denoted by lower-case and upper-case boldface letters. The symbol $|\cdot|$ represents the absolute value, while $\|\cdot\|$ denotes the l_2 norm. Moreover, we use $(\cdot)^H$ to denote the conjugate transpose operation. Finally, the Hadamard product is represented by $\odot$. The key symbols used in this paper are listed in Table I.

II. SYSTEM MODEL

We focus on a narrow-band XL-array downlink communication system,¹ where the BS is equipped with a dense uniform linear array (ULA) with N antennas. In this section,

¹The proposed beam training scheme can be readily extended to the wide-band communication systems where TTD circuits can be applied to avoid beam-squint effect.

the near-field channel and signal model for the ULA are introduced.

A. Near-Field Channel Model

We assume that the dense ULA is situated at the y -axis and centered at the origin. Specifically, each antenna of XL-array is located at $(0, nd_0)$, where $n \in \mathcal{N} \triangleq \{0, \pm 1, \dots, \pm \frac{N-1}{2}\}$ and d_0 respectively denote the antenna index and inter spacing. For the dense ULA, we have $d_0 = \frac{\lambda}{2}$, where λ represents the carrier wavelength. Moreover, the user is assumed to be located in the Fresnel region of the XL-array where the BS-user range r_0 is larger than $Z_F = \max\{d_R, 1.2D\}$ and smaller than the Rayleigh distance $Z_R = \frac{2D^2}{\lambda}$ with $D = (N-1)d_0$ denoting the array aperture. Moreover, $d_R = \sqrt{\frac{D^3}{\lambda_0}}$ denotes the Fresnel distance, where the Fresnel approximation is proven to be accurate [33]. For example, when $N = 257$ and $f = 30$ GHz, the Rayleigh distance is approximately 328 m, which makes the user more likely to be located in the near-field region. Then the general multi-path channel from the XL-array to the user can be modeled by the uniform spherical wave (USW) as [19]

$$\mathbf{h}_{\text{near}}^H = \sqrt{N}\beta \mathbf{b}^H(r_0, \theta_0) + \sum_{\ell=1}^L \sqrt{\frac{N}{L}} \beta_\ell \mathbf{b}^H(\bar{r}_\ell, \bar{\theta}_\ell), \quad (1)$$

which includes one LoS path $\mathbf{h}_{\text{LoS}}^H$ and L non-LoS (NLoS) paths. Herein, the parameters r_0 ($\bar{r}_\ell$) and θ_0 ($\bar{\theta}_\ell$) represent the range and spatial angle of the LoS (ℓ -th NLoS) signal path. Moreover, the parameters β and β_ℓ denote the LoS path and ℓ -th NLoS path gain, respectively. Mathematically, β can be modeled as [34]

$$\beta = \sqrt{\frac{\kappa}{\kappa+1}} \frac{\sqrt{\beta_0}}{r_0} e^{-\frac{j2\pi r_0}{\lambda}}, \quad (2)$$

where κ and β_0 represent the Rician factor and reference channel gain at a range of 1 m, respectively. Moreover, β_ℓ is assumed to follow the Gaussian distribution, i.e., $\beta_\ell \sim \mathcal{CN}(0, \sigma_0^2)$ where $\sigma_0 = \sqrt{\frac{1}{\kappa+1}} \frac{\sqrt{\beta_0}}{r_0}$.

In this paper, we mainly consider the near-field communication scenarios in high-frequency bands such as millimeter-wave (mmWave) and even terahertz (THz). In these scenarios, the NLoS channel paths exhibit negligible power owing to the severe path-loss and shadowing effects [35]. Therefore, we only consider the LoS channel and the BS-user channel can be approximated as $\mathbf{h}_{\text{near}}^H \approx \mathbf{h}_{\text{LoS}}^H$.² Based on USW model, the near-field LoS channel from BS→user can be modeled as [26]

$$\mathbf{h}_{\text{near}}^H \approx \sqrt{N}\beta \mathbf{b}^H(r_0, \theta_0), \quad (3)$$

where $\mathbf{b}^H(r_0, \theta_0)$ denotes the near-field channel steering vector, defined as [36]

$$[\mathbf{b}^H(r_0, \theta_0)]_n = \frac{1}{\sqrt{N}} e^{-\frac{j2\pi r_n}{\lambda}}, \quad \forall n \in \mathcal{N}, \quad (4)$$

with $r_n = \sqrt{r_0^2 + n^2 d_0^2 - 2r_0 \theta_0 n d_0}$ denoting range between the n -th antenna and the user. Moreover, $\theta_0 = \cos \phi_0 \in [-1, 1]$

represents the spatial angle at the BS, with ϕ_0 denoting the physical angle-of-departure (AoD) from the BS center to the user. Further, by means of Fresnel approximation, r_n can be approximated as

$$r_n \approx r_0 - n d_0 \theta_0 + \frac{n^2 d_0^2 (1 - \theta_0^2)}{2r_0}, \quad (5)$$

which is shown to be accurate in [3].

B. Signal Model

Let $x \in \mathbb{C}$ denote the transmitted symbol by the BS with unit power and $\bar{\mathbf{w}} \in \mathbb{C}^{N \times 1}$ represent the beamforming vector [37]. Then the received signal at the user is given by

$$y(\bar{\mathbf{w}}) = \sqrt{N}\beta \sqrt{P_{\text{tot}}} \mathbf{b}^H(r_0, \theta_0) \bar{\mathbf{w}} x + z, \quad (6)$$

where z is the received additive white Gaussian noise (AWGN) and $z \sim \mathcal{CN}(0, \sigma^2)$. Moreover, P_{tot} denotes the total transmit power of the BS. Then, the achievable rate in bits/second/hertz (bps/Hz) is given by

$$R = \log_2 \left(1 + \frac{P_{\text{tot}} N |\beta|^2 |\mathbf{b}^H(r_0, \theta_0) \bar{\mathbf{w}}|^2}{\sigma^2} \right). \quad (7)$$

III. BENCHMARK BEAM TRAINING SCHEMES

In this section, two benchmark beam training schemes and their drawbacks are presented.

A. 2D Exhaustive-Search Beam Training Method

The authors in [25] proposed a *polar-domain* codebook, each steering a focusing beam to a specific location. In particular, the angular domain is uniformly sampled, while the range domain is *non-uniformly* sampled. Specifically, the polar-domain codebook is given by

$$\bar{\mathcal{W}}_{\text{Pol}} = \{\bar{\mathcal{W}}_1, \dots, \bar{\mathcal{W}}_{\bar{n}}, \dots, \bar{\mathcal{W}}_N\}, \quad (8)$$

where $\bar{\mathcal{W}}_{\bar{n}} = \{\bar{\mathbf{w}}_{\bar{n},1}, \dots, \bar{\mathbf{w}}_{\bar{n},v}, \dots, \bar{\mathbf{w}}_{\bar{n},V}\}$ denotes the sub-codebook steering V beams towards the angles $\theta_{\bar{n}} = \frac{2\bar{n}-N+1}{N}, \forall \bar{n} \in \bar{\mathcal{N}} \triangleq \{1, 2, \dots, N\}$. Mathematically, we have $\bar{\mathbf{w}}_{\bar{n},v} = \mathbf{b}(r_{\bar{n},v}, \theta_{\bar{n}})$ where $r_{\bar{n},v} = \frac{1}{\alpha_\Delta} (1 - \theta_{\bar{n}}^2)$, $\forall v \in \mathcal{V} \triangleq \{1, 2, 3, \dots, V\}$ with $\alpha_\Delta \triangleq \frac{N^2 d_0^2}{2\lambda \beta_\Delta^2}$. Moreover, β_Δ is a constant corresponding to the quantization loss in the range domain [25]. Given the polar-domain codebook $\bar{\mathcal{W}}_{\text{Pol}}$, a *two-dimensional exhaustive search* method can be directly applied in both the angular and range domains to search the best codeword, which yields the maximum received SNR at the user. The beam training overhead of this exhaustive-search beam training method is $T^{(\text{ex})} = NV$, which is proportional to the product of the number of antennas and range samples. When the number of antenna is large, the beam training overhead is unaffordable.

B. Two-Phase Near-Field Beam Training

To further reduce the beam training overhead, the authors in [26] proposed a two-phase near-field beam training method, which explored the so-called energy-spread phenomenon. Specifically, when the far-field DFT codebook is used for the angular sweeping in the near-field, it is observed that the user

²The case where comparable multi-path components exist is more complicated and discussions are provided in the Section V-D. We will evaluate the Rician fading channel in the simulation results.

angle approximated lies in the middle of an angular support region, which is to be defined in Section IV. Mathematically, the conventional DFT codebook is given by [38]

$$\bar{\mathbf{w}}_{\bar{n}} = \mathbf{a}(\theta_{\bar{n}}) \triangleq \frac{1}{\sqrt{N}} \left[1, e^{-j\pi\theta_{\bar{n}}}, \dots, e^{-j\pi(N-1)\theta_{\bar{n}}} \right], \quad (9)$$

where $\theta_{\bar{n}} = \frac{2\bar{n}-N+1}{N}$, $\forall \bar{n} \in \bar{\mathcal{N}}$. Given this observation, they explore the conventional DFT codebook in the first phase to perform beam sweeping, which estimates the user angle information. Then, given the candidate user angle, the polar-domain codebook in (8) is used to search the best user range in the second phase. The beam training overhead of the two-phase beam training method is $T^{(2P)} = N + KV$ with K representing the number of candidate user angles. Although this method significantly reduces the beam training overhead of the exhaustive-search method, the overhead of the two-phase beam training method is still proportional to the number of antennas, which is prohibitively high as N is sufficiently large.

To address the above issues, we propose a new near-field beam training method using a proposed sparse DFT codebook, which is equivalent to sparsely activating the dense ULA equipped by the BS, yielding extremely lower overhead as compared with various benchmark schemes.

IV. RECEIVED BEAM PATTERN OF THE SPARSE DFT CODEBOOK

In this section, we first introduce the sparse DFT codebook and then analyze its received beam pattern. Moreover, we design a periodical beam training codebook to reduce overhead and propose to activate a central subarray for resolving the angular ambiguity. The main definitions in this section are given as follows, including the received beam pattern and angular support.

Definition 1: Given a fixed near-field user located at (r_0, θ_0) and an arbitrary far-field beamforming vector $\bar{\mathbf{w}} = \mathbf{a}(\theta)$ steering the beam towards the angle θ , the received beam pattern at the user is defined as

$$f(r_0, \theta_0; \theta) \triangleq |\mathbf{b}^H(r_0, \theta_0)\mathbf{a}(\theta)|, \forall \theta. \quad (10)$$

Definition 2: Given a near-field channel response vector $\mathbf{b}^H(r_0, \theta_0)$ and a far-field beamforming vector $\bar{\mathbf{w}} = \mathbf{a}(\theta)$ with $\theta \in \mathcal{L}$, the μ -dB angular support $\mathcal{A}_{\mu}^{\mathcal{L}}(r_0, \theta_0)$ in the region $\mathcal{L}$ is defined by [26]

$$\mathcal{A}_{\mu}^{\mathcal{L}}(r_0, \theta_0) = \left\{ \theta \mid f(r_0, \theta_0; \theta) > \kappa \max_{\theta \in \mathcal{L}} f(r_0, \theta_0; \theta) \right\}, \quad (11)$$

where $\kappa = 10^{\mu/10}$. Moreover, let θ_{left} and θ_{right} be the smallest and largest angle in $\mathcal{A}_{\mu}^{\mathcal{L}}(r_0, \theta_0)$. Then, its angular support width is defined as

$$\Gamma_{\mu}^{\mathcal{L}}(r_0, \theta_0) = \theta_{\text{right}} - \theta_{\text{left}}. \quad (12)$$

A. Sparse DFT Codebook

Each column of the DFT Codebook in (9) is a far-field channel response vector. For the sparse DFT codebook, we sample each column of the original DFT codebook using an interval of

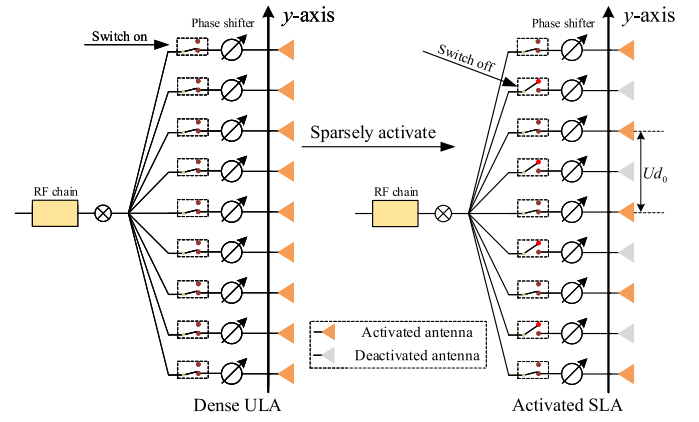


Fig. 2. Illustration of the sparse linear array and hardware structure.

U , while the positions that are not sampled are padded with zeros. As such, $\bar{n}$ -th column of the sparse DFT Codebook $\tilde{\mathbf{w}}_{\bar{n}} \in \mathbb{C}^{N \times 1}$ is given by $\tilde{\mathbf{w}}_{\bar{n}} = \bar{\mathbf{w}}_{\bar{n}} \odot \mathbf{n}(U)$, where

$$\mathbf{n}^H(U) = [1, \underbrace{0, \dots, 0}_{U-1}, 1, \dots, 1] \quad (13)$$

denotes the sampling vector. It can be verified that the number of non-zero elements in $\tilde{\mathbf{w}}_{\bar{n}}$ is $Q = \frac{N-1}{U} + 1$ (assuming Q is an integer for convenience). Moreover, we rearrange the non-zero elements in $\tilde{\mathbf{w}}_{\bar{n}}$ into a new vector $\mathbf{w}_{\bar{n}} \in \mathbb{C}^{Q \times 1}$, which is given by

$$\mathbf{w}_{\bar{n}} = \mathbf{a}_{\text{SLA}}(\theta_{\bar{n}}) = \frac{1}{\sqrt{Q}} \left[1, e^{j\pi U \theta_{\bar{n}}}, \dots, e^{j\pi(Q-1)U \theta_{\bar{n}}} \right]^H, \quad (14)$$

referred to as the *sparse far-field beamforming vector*.

It is noteworthy that this sampling method is equivalent to sparsely activating a number of antennas of the the XL-array with an interval of U , which effectively transforms the equipped ULA into an SLA with an inter-element spacing of Ud_0 [39], as illustrated in Fig. 2.³ Furthermore, for brevity, we denote the channel response vector of the effective SLA as

$$[\mathbf{b}_{\text{SLA}}^H(r_0, \theta_0)]_q = \frac{1}{\sqrt{Q}} e^{-\frac{j2\pi r_q}{\lambda}}, \forall q \in \mathcal{Q}, \quad (15)$$

where $\mathcal{Q} \triangleq \{0, \pm 1, \dots, \pm \frac{Q-1}{2}\}$ denotes the set of the SLA antenna index and $r_q = \sqrt{r_0^2 - 2qUd_0r_0\theta_0 + (qUd_0)^2}$ represents the range between the user and q -th antenna of the SLA. Similar to (5), r_q can be approximated as $r_q \approx r - qUd_0\theta_0 + \frac{q^2(Ud_0)^2(1-\theta_0^2)}{2r_0}$ with Fresnel approximation. Then, the received signal with beamforming vector $\mathbf{w}_{\bar{n}}$ at the user is rewritten as

$$y(\mathbf{w}_{\bar{n}}) = \sqrt{Q}\beta\sqrt{P_{\text{tot}}}\mathbf{b}_{\text{SLA}}^H(r_0, \theta_0)\mathbf{w}_{\bar{n}}x + z. \quad (16)$$

Moreover, the received beam pattern at the user in (10) can be rewritten as

$$\begin{aligned} f(r_0, \theta_0; \theta_{\bar{n}}) &= |\mathbf{b}^H(r_0, \theta_0)\tilde{\mathbf{w}}_{\bar{n}}| = |\mathbf{b}_{\text{SLA}}^H(r_0, \theta_0)\mathbf{w}_{\bar{n}}| \\ &= |\mathbf{b}_{\text{SLA}}^H(r_0, \theta_0)\mathbf{a}_{\text{SLA}}(\theta_{\bar{n}})|. \end{aligned} \quad (17)$$

³The antenna sparse activation can be achieved by a switch network. If a switch is turned off, the antenna attached to this switch is deactivated as shown in Fig. 2 [40].

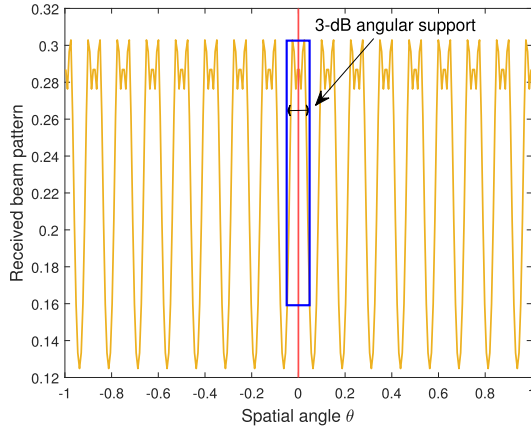


Fig. 3. Received beam pattern, where $N = 257$, $U = 16$ and $f = 30$ GHz. The actual user angle is $\theta_0 = 0$ marked by the red line, while the 3-dB angular support is marked by the blue box.

B. Near-Field Received Beam Pattern

To obtain more insights, we first characterize the received beam pattern at the user with sparse far-field beamforming vectors spanning in the continuous spatial angles. Let $\mathbf{w}(\theta) = \mathbf{a}_{\text{SLA}}(\theta)$ represent a sparse far-field beamforming vector, for which $\theta = \theta_{\bar{n}}, \forall \bar{n} \in \bar{\mathcal{N}}$ is the discrete sampled angle. We first characterize the received beam pattern of the sparse far-field beamforming vector $\mathbf{w}$ as follows.

Lemma 1: For a sampled beamforming vector $\mathbf{w}$ parameterized by $\{Q, U\}$, the received beam pattern at the user is given by

$$\begin{aligned} f(r_0, \theta_0; \theta) &\stackrel{(a_1)}{\approx} \frac{1}{Q} \left| \sum_{q \in \mathcal{Q}} \exp \left(\underbrace{j\pi q U \Delta}_{B_1} + \underbrace{j\frac{\pi}{\lambda} q^2 (U d_0)^2 \frac{1 - \theta_0^2}{r_0}}_{B_2} \right) \right| \\ &\triangleq \hat{f}(r_0, \theta_0; \theta), \end{aligned} \quad (18)$$

where $\Delta \triangleq \theta - \theta_0$ and (a_1) is due to the Fresnel approximation and shown to be accurate in [41].

Proposition 1 (The periodicity of θ): $f(r_0, \theta_0; \theta)$ is a periodic function of θ with the period of $\frac{2}{U}$. Mathematically, we have

$$f(r_0, \theta_0; \theta) = f\left(r_0, \theta_0; \theta + \frac{2k}{U}\right), \forall k \in \mathbb{Z}. \quad (19)$$

Proof: For an arbitrary integer k and $\mathbf{a}(\theta + \frac{2k}{U})$, we have

$$\begin{aligned} [\mathbf{a}(\theta + 2k/U)]_q &= \exp(j\pi q U (\theta + 2k/U)) \\ &= \exp(j\pi q U \theta) \exp(j2\pi k q U) \\ &= \exp(j\pi q U \theta) = [\mathbf{a}(\theta)]_q. \end{aligned}$$

Hence, we have $f(r_0, \theta_0; \theta) = f(r_0, \theta_0; \theta + k\frac{2}{U})$ and thus complete the proof. $\square$

In Fig. 3, we plot the received beam pattern versus the spatial angle θ , where the periodicity of θ is exhibited. Moreover, it is observed that the user angle is located in the middle of the angular support $\mathcal{A}_\mu^{\mathcal{L}_0}(r_0, \theta_0)$, where $\mathcal{L}_0 = [\theta_0 - 1/U, \theta_0 + 1/U)$

(a period $\frac{2}{U}$). Then, in the following, we prove that this observation holds for arbitrary user location (r_0, θ_0) .

Due to the periodicity of θ , we only need to focus on the region $\mathcal{L}_0$, i.e., $\Delta \in [-1/U, 1/U)$, which inexplicitly contains useful user angle information. Given $\Delta \in [-1/U, 1/U)$, (18) can be approximated as follows.

Lemma 2: When $\Delta \in [-1/U, 1/U)$, the received beam pattern $\hat{f}(r_0, \theta_0; \theta)$ in (18) can be approximated as

$$\hat{f}(r_0, \theta_0; \theta) \approx |G(\beta_1, \beta_2)|. \quad (20)$$

Specifically, we have

$$G(\beta_1, \beta_2) \triangleq \frac{\widehat{C}(\beta_1, \beta_2) + j(\widehat{S}(\beta_1, \beta_2))}{2\beta_2}, \quad (21)$$

where $\widehat{C}(\beta_1, \beta_2) \triangleq C(\beta_1 + \beta_2) - C(\beta_1 - \beta_2)$ and $\widehat{S}(\beta_1, \beta_2) \triangleq S(\beta_1 + \beta_2) - S(\beta_1 - \beta_2)$. Further, $C(x)$ and $S(x)$ are Fresnel integrals, which are given by

$$C(x) = \int_0^x \cos(\frac{\pi}{2} t^2) dt, S(x) = \int_0^x \sin(\frac{\pi}{2} t^2) dt.$$

Moreover, β_1 and β_2 are given by

$$\beta_1 = \Delta \sqrt{\frac{r_0}{d_0(1 - \theta_0^2)}}, \quad \beta_2 = \frac{QU}{2} \sqrt{\frac{d_0(1 - \theta_0^2)}{r_0}}. \quad (22)$$

Proof: Please refer to Appendix A. $\square$

As observed from (22), we have

$$\beta_1 \beta_2 = \frac{QU\Delta}{2} \approx \frac{D}{2d_0} \Delta.$$

Hence, the function $G(\cdot)$ can be rewritten as a function of $\{\Delta, \beta_2\}$. It is observed that if the array aperture is fixed, β_2 is only determined by the user location (r_0, θ_0) , which indicates that each user location corresponds to a specific β_2 . Furthermore, as the user moves farther from the BS (consequently, the user is more likely to be located in the far-field region), the value of β_2 decreases. Supposing that $N = 257$, $U = 16$ and $f = 30$ GHz and the user is located at the Fresnel and focusing region ($\theta_0 = 0$), we have $\beta_2 \in [1.68, 3.57]$. In Fig. 4, we numerically show the function of $G(\cdot)$. Importantly, for any near-field users, the so-called energy-spread phenomenon [25] is still observed. Moreover, due to the period of θ , the energy-spread effects exist in each period interval $(\frac{2}{U})$. Then, two key observations are obtained.

Observation 1: In Figs. 3 and 4, the energy-spread effects exhibit in the whole radiating near-field region, and the received beam pattern with sampled beamforming vector $\mathbf{w}$ contains useful user information:

- 1) **User angle information:** It is observed that the actual user angle approximately locates in the middle of the 3-dB angular support in one period where $\theta \in \mathcal{L}_0 = [\theta_0 - 1/U, \theta_0 + 1/U)$. Mathematically, we have

$$\theta_0 \approx \text{Med}(\mathcal{A}_3^{\mathcal{L}_0}(\theta_0, r_0)), \quad (23)$$

where the symbol $\text{Med}(\mathcal{L}_0)$ represents the middle of the range $\mathcal{L}_0$. Moreover, we define u -th angular support as $\mathcal{A}_3^{\mathcal{L}_u}(\theta_0, r_0)$, where $\mathcal{L}_u = \mathcal{L}_0 + \frac{2u}{U}$, $\forall u \in \mathcal{U} \triangleq \{\pm 1, \pm 2, \dots, \pm(U-1)\} \cap \{k|\theta_0 + \frac{2k}{U} \in [-1, 1]\}$. If we

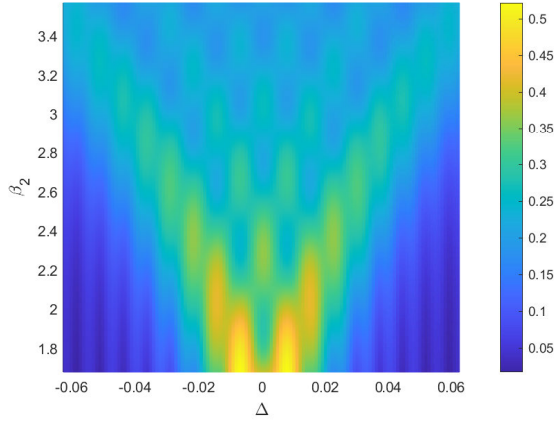


Fig. 4. Received beam pattern in one period, where $N = 257$, $U = 16$ and $f = 30$ GHz.

can employ the middle of u -th angular support denoted by $\theta_u \approx \text{Med}(\mathcal{A}_3^u(\theta_0, r_0))$, the BS can infer that the user angle is among the following candidate angles $\theta_u \approx \text{Med}(\mathcal{A}_3^u(\theta_0, r_0))$, where $\mathcal{L}_c = \mathcal{L}_u + \frac{2c}{U}$, $\forall c \in \mathcal{U}$, due to the period of $\frac{2}{U}$.

- 2) Although energy spread effects can provide actual user angle information, it is worth noting that when the user is located near the boundary of Fresnel region (in Fig. 4, the boundary corresponds to $\beta_2 = 3.57$ and $r_0 = 7.42\text{m}$), the angular support is distorted, which affects the accuracy of the beam training results. Moreover, by taking the noise and power fluctuation into account, the actual user angle may slightly deviate from the middle of the angular support.

Observation 1 indicates that the actual user angle can be estimated by finding the middle of the angular support in a period. In other words, the sparse DFT codebook within the angle range of a period (for example $\theta \in [-1/U, 1/U]$) contained all information for which the BS can infer U candidate angles for actual user angle.

V. PROPOSED BEAM TRAINING SCHEME ENABLED BY SPARSE DFT CODEBOOK

In this section, we propose a three-phase near-field beam training method enabled by the sparse DFT codebook. Then, an optimization problem is formulated to minimize the beam training overhead.

A. Phase 1: Beam Sweeping With the Sparse DFT Codebook

In Section IV-B, we show that the received beam pattern with the sparse DFT codebook exhibits a period of $2/U$ and the user angle information can be inferred from the middle of the angular support within the region $\mathcal{L}_0$.

This motivates us to perform angular sweeping in a period of θ (steering beams varying from $-1/U$ to $1/U$) to estimate the potential user angles in the first phase, thereby further decreasing the beam training overhead. Considering that the beam width of the sparse far-field beamforming vector (14) is

$\frac{4}{QU}$ [42], we sample the angular space as

$$\theta_s = \frac{2s - QU - 1}{QU}, s = 1, \dots, QU.$$

Then, the sparse DFT codebook in the first phase to perform the angular sweeping is given by

$$\mathcal{W}_{\text{DFT}} = \{\mathbf{w}_{\frac{QU-Q+1}{2}}, \dots, \mathbf{w}_g, \dots, \mathbf{w}_{\frac{QU+Q-1}{2}}\}, \quad (24)$$

where $\mathbf{w}_g = \mathbf{a}_{\text{SLA}}(\theta_g)$, $\theta_g = \frac{2g - QU - 1}{QU}$ and the index $g \in \mathcal{G} \triangleq \{\frac{QU-Q+1}{2}, \dots, \frac{QU+Q-1}{2}\}$.

Then, the BS sequentially transmits Q pilot symbols with the sparse DFT codebook in (24), while it tunes beam angles varying from $-1/U$ to $1/U$ as illustrated in Fig. 5(a). For each codeword, the received signal power at the user is given by

$$p(\mathbf{w}_g) = |\sqrt{Q}\beta \mathbf{b}^H(r_0, \theta_0) \mathbf{w}_g x + z|^2, \forall g \in \mathcal{G}. \quad (25)$$

However, it is worth noting that when we transmit sparse far-field beamforming vectors varying from $-1/U$ to $1/U$, the received beam pattern may exhibit a *shifted* angular support within two periods instead of a whole angular support in one period as illustrated in Fig. 6. To obtain a regular angular support, the user needs to perform the received-beam-pattern shifting based on the index of the codeword $\mathbf{w}_g$ with the lowest power. We denote the codeword with the lowest power as $\mathbf{w}_\ell$ corresponding to the angle θ_ℓ . Specifically, we shift the angle larger than θ_ℓ by one period ($2/U$) as illustrated in Fig. 6. Then, the shifted indices of the codewords are arranged in a vector $\mathbf{s} = [\ell - Q + 1, \dots, \ell - Q + 2, \dots, \ell]$. Moreover, the equivalent received power of the shifted codewords can be recast as

$$\mathbf{P} = \{p(\mathbf{w}_{\ell+1}), p(\mathbf{w}_{\ell+2}), \dots, p(\mathbf{w}_{(QU+Q-1)/2}), p(\mathbf{w}_{(QU-Q+1)/2}), p(\mathbf{w}_{(QU-Q+3)/2}), \dots, p(\mathbf{w}_\ell)\}.$$

As such, we obtain a complete and regular angular support, which involves the user's angle information based on **Observation 1**. Specifically, the indices of shifted codewords with significantly high power received by the users are given by

$$\mathbf{S} = \{\mathbf{s}_\delta | p(\mathbf{w}_{\mathbf{s}_\delta}) > \kappa \max \mathbf{P}\}, \quad (26)$$

where $\kappa \approx 0.5$. Then, the estimated angle is given by $\theta_{\tilde{s}} = \frac{2\tilde{s} - QU - 1}{QU}$, where $\tilde{s} = \text{Med}(\mathbf{S})$ and the corresponding codeword is $\mathbf{w}_{\tilde{s}}$. Then, according to the periodicity, the BS can infer U candidate user angles, which are given by

$$\theta_c = \theta_{\tilde{s}} + \frac{2u}{U}, \forall u \in \mathcal{U}. \quad (27)$$

B. Phase 2: Angular Ambiguity Elimination

In the second phase, we propose an efficient method to resolve the angular ambiguity. The key idea is to utilize a central subarray to sequentially examine the candidate angles in (27). Specifically, we activate a central subarray with M antennas to eliminate the angular ambiguity, for which the codebook $\mathcal{V}_{\text{Sub}}$ is presented as follows

$$\mathcal{V}_{\text{Sub}} = \{\mathbf{v}_1, \dots, \mathbf{v}_s, \dots, \mathbf{v}_{QU}\}, \quad (28)$$

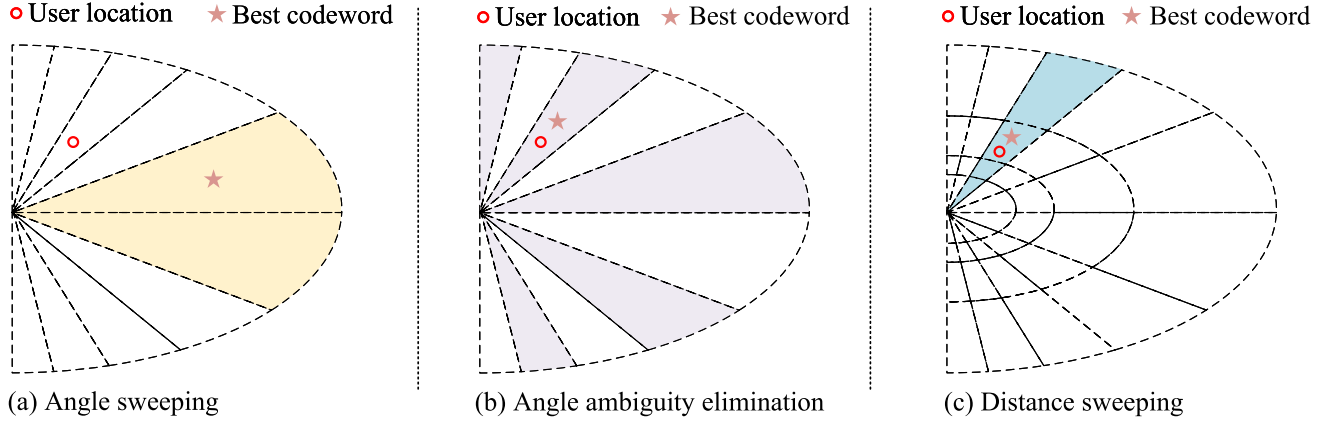
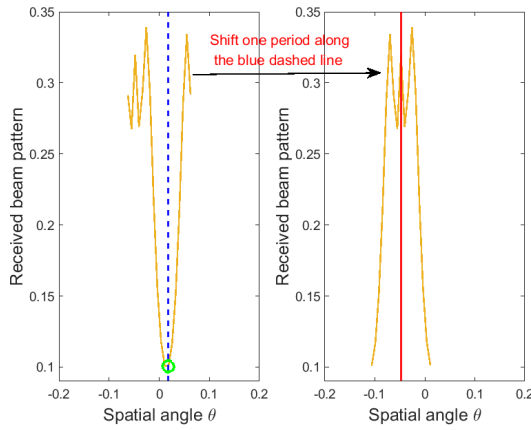


Fig. 5. Illustration of the proposed three-phase training scheme based on the sparse DFT codebook.


 Fig. 6. The shift of the received beam pattern with the sparse DFT codebook in a period, where $N = 257$, $U = 16$ and $f = 30$ GHz. The actual user angle is $\theta_0 = 0.2$.

where $\mathbf{v}_s$ is given by

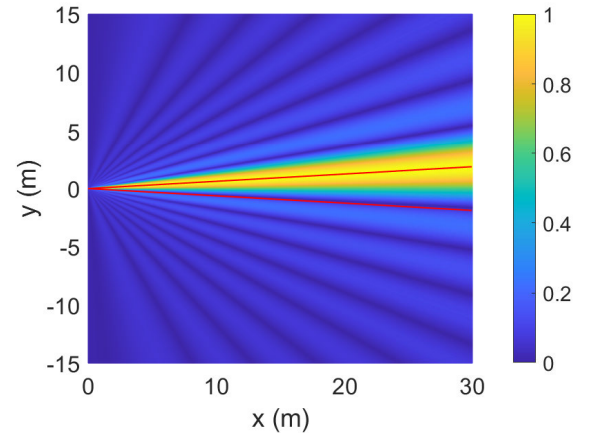
$$\mathbf{v}_s = \frac{1}{\sqrt{M}} \left[\underbrace{0, \dots, 0}_{\frac{N-M}{2}}, e^{j\pi \frac{N-M}{2} \theta_s}, \dots, e^{j\pi \frac{N+M-2}{2} \theta_s}, \underbrace{0, \dots, 0}_{\frac{N-M}{2}} \right]^H. \quad (29)$$

It is noteworthy that there are two criteria for ensuring the effectiveness of angular ambiguity elimination. The first one is that the user should be located in the far-field region of the activated central subarray to avoid the energy-spread effect. Moreover, considering that the interval between two adjacent candidate angles is $\frac{2}{U}$, extra angular ambiguity will be introduced if the beam width of the central subarray is wider than $\frac{4}{U}$. Therefore, the second criterion is that the beam width of the central subarray is required to be smaller than $\frac{4}{U}$, which ensures no interference between two adjacent candidate angles.

To satisfy the first criterion, we set the Rayleigh distance of the subarray to be no large than the minimal user range $r_{\min}$, which is given by

$$\frac{2M^2 d_0^2}{\lambda} \leq r_{\min}. \quad (30)$$

Then, (30) can be simplified as $M \leq \sqrt{\frac{r_{\min}}{d_0}}$.


 Fig. 7. Illustration of angle ambiguity eliminated by the central subarray, where $N = 257$, $M = 17$ and $f = 30$ GHz. The beam is steered towards an candidate angle $1/16$. The red lines are two adjacent candidate user angles $-1/16$ and $1/16$.

For the second criterion, we have

$$\frac{4}{M} \leq \frac{4}{U}, \quad (31)$$

which can be simplified to $M \geq U$. Therefore, the number of antennas of the central subarray needs to satisfy

$$U \leq M \leq \sqrt{\frac{r_{\min}}{d_0}}. \quad (32)$$

Moreover, from (32), the activation interval U has the following constraint $U \leq \sqrt{\frac{r_{\min}}{d_0}}$. For example, given a setup where $N = 257$, $U = 16$ and $r_{\min} = 1.2D = 1.2$ m, we have $16 \leq M \leq 17.5$. In Fig. 7, we plot the beam width of the subarray with $M = 17$. It is observed that two adjacent candidate user angles have no considerable mutual interference, which indicates the effectiveness of angular ambiguity elimination.

Based on the above, the BS activates a central subarray comprising M antennas satisfying (32) to resolve the angular ambiguity. For the candidate user angles $\theta_s + u \frac{2}{U}, \forall u \in \mathcal{U}$ in (27), the codewords chosen from (28) to resolve the angular ambiguity are $\mathbf{v}_{s+uQ}, \forall u \in \mathcal{U}$. Then, the BS sequentially transmits U pilot symbols with beamforming vectors $\{\mathbf{v}_{s+uQ}\}$

as illustrated in Fig. 5 (b). For each codeword, the received signal power at the user in the second phase is given by

$$p(\mathbf{v}_{\bar{s}+uQ}) = |\sqrt{M}\beta\mathbf{b}^H(r_0, \theta_0)\mathbf{v}_{\bar{s}+uQ}x + z|^2, \quad \forall u \in \mathcal{U}. \quad (33)$$

Then, we can obtain the optimal codeword $\mathbf{v}_{s^*}$ via simple comparisons of received signal power in (33), which is given by

$$s^* = \arg \max_{u \in \mathcal{U}} p(\mathbf{v}_{\bar{s}+uQ}). \quad (34)$$

Then, the estimated user angle can be obtained by $\theta_{s^*} = \frac{2s^* - QU - 1}{QU}$.

C. Phase 3: Beam Sweeping With Polar-Domain Codebook

Once we determine the user angle θ_{s^*} in the second phase, we can use the polar-domain codebook to perform the range domain sweeping, achieving beamfocusing gain in the near-field. The polar-domain codebook utilized in the third phase is given by

$$\bar{\mathcal{X}}_{\text{Pol}} = \{\mathcal{X}_1, \dots, \mathcal{X}_s, \dots, \mathcal{X}_{QU}\}, \quad (35)$$

where $\mathcal{X}_s = \{\mathbf{x}_{s,1}, \dots, \mathbf{x}_{s,v}, \dots, \mathbf{x}_{s,V}\}$ and $\mathbf{x}_{s,v} = \mathbf{b}^H(r_{s,v}, \theta_s)$ with $r_{s,v} = \frac{1}{v}\alpha_\Delta(1 - \theta_s^2)$ [25]. Moreover, V denotes the number of range samples.

Specifically, the codewords employed in the third phase are given by $\mathcal{X}_{s^*} = \{\mathbf{x}_{s^*,1}, \dots, \mathbf{x}_{s^*,v}, \dots, \mathbf{x}_{s^*,V}\}$. Then, the BS activates the whole XL-array and sequentially transmits training V symbols in the estimated user angle θ_{s^*} with codewords $\mathcal{X}_{s^*}$ as illustrated in Fig 5(c). For each codeword, the received signal power at the user is given by

$$p(\mathbf{x}_{s^*,v}) = |\sqrt{N}\beta\mathbf{b}^H(r_0, \theta_0)\mathbf{x}_{s^*,v}x + z|^2, \quad \forall v \in \mathcal{V}. \quad (36)$$

Subsequently, the optimal codeword is determined through straightforward comparisons of received signal power in (36) and the best codeword index is given by

$$v^* = \arg \max_{v \in \mathcal{V}} p(\mathbf{x}_{s^*,v}). \quad (37)$$

Therefore, we can obtain the user location $(r_{s^*,v^*}, \theta_{s^*})$ with $r_{s^*,v^*} = \frac{1}{v^*}\alpha_\Delta(1 - \theta_{s^*}^2)$.

The detailed procedures of the proposed three-phase beam training method are summarized in Algorithm 1.

D. Discussions and Extensions

1) *Beam Training Overhead*: In the first phase, the overhead is $Q = \frac{N-1}{U} + 1$. Moreover, the overhead of the second and third phases are U and V , respectively. Finally, the overall overhead of the proposed beam training scheme is $T^{(3P)} = Q + U + V$. Given $Q = \frac{N-1}{U} + 1 \in \mathbb{Z}^+$ with $\mathbb{Z}^+$ denoting the positive integer set, the beam training overhead of the proposed method enabled by the sparse DFT codebook can be recast as

$$T^{(3P)} = \frac{N-1}{U} + U + V + 1. \quad (38)$$

It can be easily obtained from (38) that as U increases, the overhead during the first phase decreases, while it increases

Algorithm 1 Proposed Three-Phase Beam Training Method

- 1: **Phase 1: Angular sweeping in the subspace.**
- 2: Use the sparse DFT codebook $\mathcal{W}_{\text{DFT}}$ for the beam sweeping in the angular subspace $[-1/U, 1/U]$.
- 3: Perform the received-beam-pattern shifting to obtain a regular angular support $\mathcal{P}$.
- 4: Obtain U candidate user angles $\theta_c = \theta_s + \frac{2u}{U}, \forall u \in \mathcal{U}$ according to the middle of the angular support $\mathcal{P}$.
- 5: **Phase 2: Resolve angular ambiguity.**
- 6: Activate a central subarray with M antennas and use codebook $\mathcal{V}_{\text{Sub}}$ to examine U candidate angles estimated in the first phase.
- 7: Obtain the best user angle θ_{s^*} according to the highest power received at the user with respective to $p(\mathbf{v}_{\bar{s}+uQ})$.
- 8: **Phase 3: Range sweeping.**
- 9: Use polar-domain codebook $\bar{\mathcal{X}}_{\text{Pol}}$ with codewords $\mathcal{X}_{s^*}$ to sweep the range domain in the user angle θ_{s^*} .
- 10: Obtain the best user range r_{s^*,v^*} according to the highest power received at the user with respective to $p(\mathbf{x}_{s^*,v})$.

during the subsequent phase, and vice versa. Thus, there is a fundamental trade-off between the first and second phase of the beam training method depending on the value of U . Then, we aim to minimize the beam training overhead $T^{(3P)}$ and the optimization problem is formulated as

$$\begin{aligned} (\mathbf{P1}): \quad & \min_U F(U) = \frac{N-1}{U} + U + V + 1 \\ & \text{s.t. } U \leq \sqrt{\frac{r_{\min}}{d_0}}, \\ & U \in \mathcal{F} \triangleq \{x \mid \frac{N-1}{x} + 1 \in \mathbb{Z}^+, x \in \mathbb{Z}^+\} \end{aligned} \quad (39a)$$

$$. \quad (39b)$$

If we remove the integer constraint (39b), (P1) is a convex problem. Then, (P1) can be rewritten as

$$\begin{aligned} (\mathbf{P2}): \quad & \min_U F(U) = \frac{N-1}{U} + U + V + 1 \\ & \text{s.t. } U \leq \sqrt{\frac{r_{\min}}{d_0}}. \end{aligned} \quad (40a)$$

(P2) is an convex optimization problem, whose optimal solution can be easily obtained as follows due to convexity.

Lemma 3: The optimal solution to (P2) is given by $U_{P_2}^* = \sqrt{(N-1)}$. Moreover, the overhead of the proposed method is $T^{(3P)} = 2\sqrt{(N-1)} + V + 1$.

Proof: It is observed that $\frac{N-1}{U} + U + V + 1 \geq 2\sqrt{N-1} + V + 1$, where the equal holds with $U = \sqrt{(N-1)}$. Moreover, considering that $r_{\min} \geq Z_F \geq 1.2D$, the constraint (40a) can be rewritten as $U \leq \sqrt{1.2(N-1)}$. Then, we have $U = \sqrt{(N-1)} \leq \sqrt{1.2(N-1)}$ and the optimal solution to (P2) is $U_{P_2}^* = \sqrt{(N-1)}$ with the beam training overhead $T^{(3P)} = 2\sqrt{(N-1)} + V + 1$. $\square$

When the integer constraint (39b) is taken into account, a suboptimal solution to (P1) is given by

$$\hat{U} = \arg \min_{\mathcal{F}} \{|U_{P_2}^* - f|, f \in \mathcal{F}\}. \quad (41)$$

Next, we present an example to illustrate the low overhead feature of the proposed beam training method. We consider a setup with $N = 1025$ and $V = 5$. Then, $U = 32$ can minimize $T^{(3P)}$ according to the optimal solution to (P1). The beam training overhead of the proposed three-phase beam training scheme in this setup is $T^{(3P)} = 2\sqrt{N-1} + V + 1 = 64 + 5 + 1 = 70$, which is significantly smaller than that of the exhaustive-search method ($T^{(EX)} = 5280$) and the two-phase near-field beam training method ($T^{(2P)} = 1061$).

2) *Extension to Multi-User Scenario*: We assume that the BS serves I users in the near-field regions and we use the subscript i to denote the i -th user, $i \in \mathcal{I} \triangleq \{1, 2, \dots, I\}$. For the multi-user scenario, the first phase is beam sweeping with the sparse DFT codebook, which is the same as the single user scenario in Section V-A. However, in the second phase, If we still resolve the angular ambiguity for I users point by point, the beam training overhead becomes UI , which increases with the number of users. Hence, we perform the wide beam sweeping to eliminate the angular ambiguity for each user. Specifically, we activate a central subarray consisting of M antennas and the sampling angles are $\theta_m = \frac{2m-M-1}{M}$, $m \in \mathcal{M} \triangleq \{1, 2, \dots, M\}$. We denote the codeword index with the largest received power for each user by m_i^* , corresponding to the sampling angle $\theta_{m_i^*} = \frac{2m_i^*-M-1}{M}$. Then, the actual user angle can be estimated as follows

$$\theta_i^* = \arg \min_{\theta_i} |\theta_{m_i^*} - \theta_i|. \quad (42)$$

Moreover, to avoid determination interference, we should ensure that one wide beam can not cover two adjacent user angle and we have $\frac{4}{M} \leq \frac{2}{U}$, i.e., $M \geq 2U$. In the third phase, the procedure is similar to the single user scenario, which searches the best range parameter for each user.

Remark 1 (Estimation error): From the above analysis, the accuracy of the proposed beam training scheme enabled by sparse DFT codebook is mainly dependent on two factors. First, the sampling resolution is a key factor because the proposed beam training scheme is an on-grid channel estimation method. Therefore, when the sampling interval is small enough, the performance of proposed three-phase beam training scheme approaches that of the optimal beamformer without noise taking into account. Second, noise is another key issue. The estimated angle is derived from the received signal power in different time, consequently influenced by the received SNR. Hence, a higher SNR is expected to achieve more accurate angle estimation, which will be numerically verified in Section VI.

Remark 2 (Improved scheme: Middle- K angle selection): Due to the power fluctuation [26], the middle angle of the angular support may not be accurate. To improve the estimation accuracy, we can select the *middle- K angles* of the angular support instead of selecting one potential angle in the first procedure. Considering that the beam width of the wide beam generated by the subarray is $\frac{2}{M}$, which is usually larger than the width of K angles ($\frac{2K}{QU}$) with small K [26] in the first phase, the wide beam can cover the entire adjacent K angles and we only need to perform the Middle- K angle selection in the third phase. Specifically, in the third phase,

we should perform the beam sweeping in the range domain in the K potential angles $\theta_{\bar{s}_k}$, $k = 1, \dots, K$ to determine an optimal polar-domain codeword. The overhead of the middle- K angle scheme is given by $T_{\text{Mid-}K}^{(3P)} = \frac{N-1}{U} + U + KV$. It is noteworthy that this does not significantly increase the beam training overhead, which is still proportional to $\sqrt{N}$.

Remark 3 (Multi-path channels): With respect to multi-path channel cases, we divide the extension of the proposed algorithm into two cases according to different Rician parameters.

- **LoS-dominant channel**: When the Rician factor is large enough (or equivalently the LoS path is dominant), we can regard the NLoS components as environmental noise. Therefore, the proposed beam training scheme based on the sparse DFT codebook still holds as this method only depends on the LoS path component.
- **Comparable multi-path components**: This case is much more complicated. Considering that the accuracy depends on the periodic angular supports, NLoS components may bring about randomly overlapped received beam pattern in the angular domain, which poses challenges to the angle estimation in the first phase of our proposed beam training scheme. Therefore, the case with comparable multi-path components is left as a topic for our future works.

Remark 4: (Universal in both near- and far-field communications): The proposed beam training scheme can be applied to both near- and far-field communications. We can identify the near- or far-field user according to the angular support width [43]. Specifically, in cases where the angular support width is small (e.g., only contains one candidate angle), it signifies that the user is located in the far-field region. Then, there is no need to perform the range estimation in the third phase (see Section V-C). In other words, for far-field users, only the first two phases of the proposed beam training scheme need to be executed.

Remark 5 (Hierarchical beam training): To further reduce the beam training overhead, hierarchical beam training methods can be utilized in Phase 2. Specifically, we can activate a central *sparse* subarray with an activation-interval $\frac{U}{2}$, instead of a central dense subarray. Then, we divide the candidate users into two groups and two pilots can form beams covering the above two group angles, due to a beam period $\frac{4}{U}$ of the activated central sparse subarray. According to simple comparisons of the received power of the two pilots, we can determine $\frac{U}{2}$ candidate user angles. Subsequently, we sequentially activate the center sparse subarray with activation-intervals of $U/4$, $U/8$, and so forth, up to 1, thereby determining of a unique user angle. As such, the beam training overhead for ambiguity elimination is $\log_2 U$, which is significantly smaller than that of U in Section V-B.

VI. NUMERICAL RESULTS

Numerical results are provided to validate the effectiveness of the proposed near-field beam training scheme in this section. We first present the system parameters and benchmark schemes followed by the performance comparison under numerous setups.

A. System Setup and Benchmark Schemes

We set the system parameters as follows. We consider that the XL-array is equipped with $N = 257$ antennas and $f = 30$ GHz. The transmit power and reference channel gain at 1 m are set as $P_{\text{tol}} = 30$ dBm and $\beta_0 = (\frac{\lambda}{4\pi})^2 = -62$ dB, respectively. Moreover, the noise power is $\sigma^2 = -80$ dBm. According to (P1), the optimal activation interval is set as $U = 16$. Furthermore, the antenna number of the activated central subarray is $M = 17$. With respect to NLoS paths, we set $L = 2$ and $\kappa = 30$ dB [27], [30]. The reference SNR for a user is defined by $\gamma = \frac{NP_{\text{tol}}\beta_0}{\tau_0^2\sigma^2}$ [26]. To characterize the overhead, we assume that the total transmission time and a pilot symbol time are $T_{\text{tol}} = 0.2$ ms and $T_s = 0.1$ μ s [44], respectively. Then, the *effective rate* is defined by $R_{\text{Eff}} = (1 - \frac{T_{\text{over}}T_s}{T_{\text{tol}}})R$, where T_{over} denotes the beam training overhead of each scheme. All the numerical results are averaged over 1000 channel realizations. The following benchmark schemes are considered for performance comparison:

- **Perfect-CSI based beamforming:** This scheme assumes that the BS perfectly aligns the user near-field channel and beamforming vector is set by $\bar{\mathbf{w}} = \mathbf{b}(r_0, \theta_0)$. Obviously, this scheme is the performance upper bound for all methods.
- **Least square channel estimation:** This scheme is a classic off-grid channel estimation method where the user estimates the channels by N pilot symbols transmitted by the BS in the downlink. The estimated channel is given by $\hat{\mathbf{h}}_{\text{LS}} = (\mathbf{X}^H\mathbf{X})^{-1}\mathbf{X}^H\mathbf{y}$, where $\mathbf{X} \in \mathbb{C}^{N \times N}$ denotes the pilot matrix for each user and $\mathbf{y} = \mathbf{X}\mathbf{h} + \mathbf{z} \in \mathbb{C}^{N \times 1}$ represents the received signal vector with $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$. Moreover, $\mathbf{h}$ is shown in (1). Obviously, the pilot overhead of this scheme is $T^{(\text{LS})} = N$.
- **Exhaustive-search beam training:** This scheme is detailed in Section III-A. Due to the different angle sampling interval, the overhead of this scheme is modified by $T^{(\text{ex})} = QUV$.
- **Two-phase beam training:** This scheme is detailed in Section III-B. Due to the different angle sampling interval, the overhead of this scheme is revised by $T^{(2P)} = QU + KV$.
- **Far-field beam training based on DFT codebook:** Conventional DFT codebook is used to sweep the whole angular domain for choosing a best codeword for which the maximum received signal power is achieved at the user. The beam training overhead of this scheme is $T^{(\text{Far})} = QU$.

B. Performance Analysis

In Fig. 8, we plot the beam training overhead versus the number of antennas. It is observed that regardless of the number of antennas, the proposed beam training scheme achieves lower beam training overhead than the benchmark schemes via leveraging the novel sparse DFT codebook. Hence, the effective rate performance of the proposed beam training scheme is significantly higher than other beam training methods (see Fig. 12).

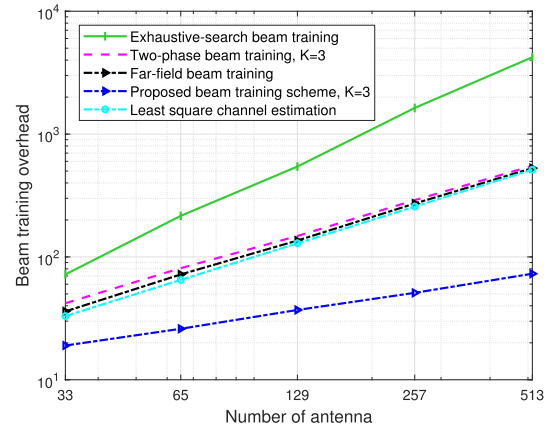


Fig. 8. Beam training overhead versus number of antennas.

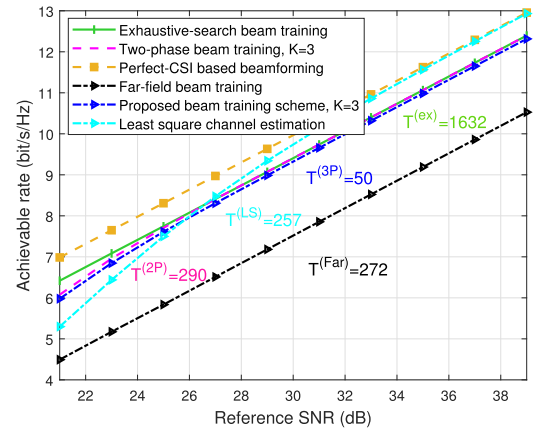


Fig. 9. Achievable rate versus SNR.

In Fig. 9, we plot the achievable rate R versus the reference SNR γ under different beam training schemes. Some key observations can be concluded as follows. First, our proposed near-field beam training scheme achieves very close performance to the two-phase beam training and exhaustive search based beam training scheme, especially in the high-SNR regime (i.e., larger than 26 dB). Second, with the decrease of the reference SNR, the achievable rate attained by off-grid channel estimation degrades more dramatically than other schemes, and becomes inferior to our proposed scheme when the reference SNR is lower than 26 dB. This is because the direct channel estimation is more sensitive to the received SNR as the XL-array beamforming vector may not be well aligned with the channel path during signaling. Finally, there is a large performance gap between the conventional far-field beam training scheme and other schemes dedicated to near-field communications, which implies that far-field beam training is no longer effective for next generation wireless systems with more antennas.

Fig. 10 shows the effect of user range on the achievable rate. It can be observed that the proposed scheme with $K = 3$ exhibits approximately the same performance as the exhaustive-search based and two-phase near-field beam training schemes for all user ranges. This is attributed to the smart design of the sparse DFT codebook. This method leads

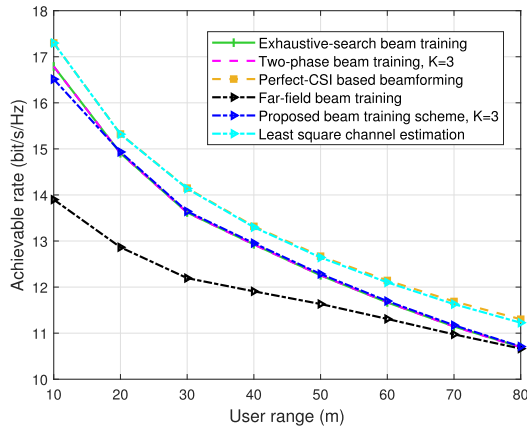
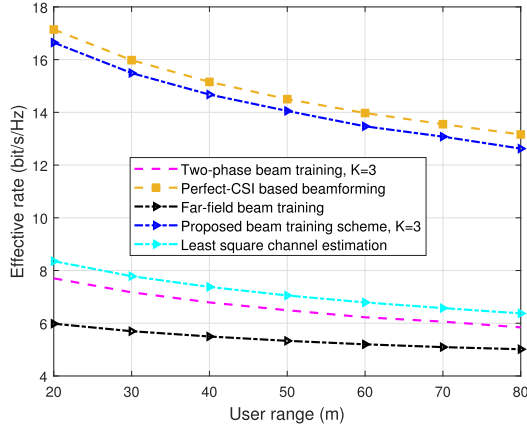


Fig. 10. Achievable rate versus user range.


 Fig. 11. Effective rate versus user range with $f = 100$ GHz.

to a periodical energy-spread effect during beam training, and hence the former key observation in [26] can be leveraged for beam training. Second, the achievable rate performance of the proposed scheme largely outperforms the far-field beam training when the user range is less than 50 m and gradually converges to that of the far-field beam training. This verifies the universality of our proposed scheme for both near-field and far-field beam training cases. In addition, to validate the effectiveness of our proposed scheme in THz bands, we plot the effective rate performance versus the user range with $f = 100$ GHz in Fig. 11. To combat with severe path loss, the total power and number of antenna are set as $P_{\text{tol}} = 40$ dBm and $N = 1025$, respectively. It is observed that the proposed three-phase beam training method achieves higher effective-rate performance as compared to benchmark schemes due to lower beam training overhead, which demonstrates the effectiveness of our proposed beam training method in the THz bands.

Then, Fig. 12 illustrates the effective rate R_{Eff} versus the reference SNR γ . Interestingly, our proposed scheme outperforms other schemes in terms of the effective rate except for the perfect-CSI based beamforming. This is because our scheme achieves close or even superior achievable rates compared with other benchmarks (as have shown in Fig. 9), but with far less training overhead. Moreover, the exhaustive search based near-field beam training scheme is not practically

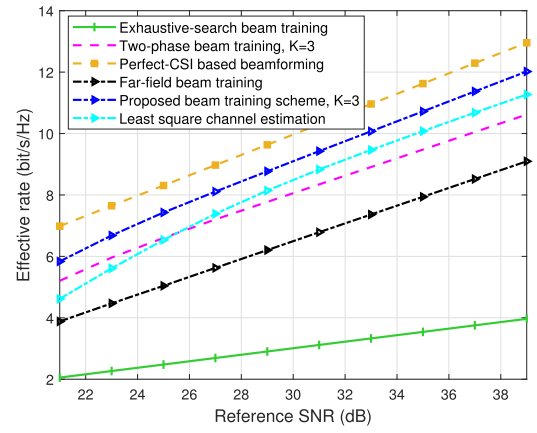


Fig. 12. Effective rate versus SNR.

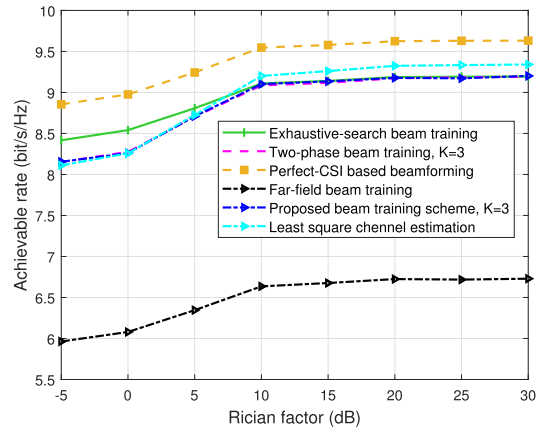


Fig. 13. Achievable rate versus Rician factor.

applicable due to small effective rate caused by large training overhead, as shown in Fig. 12.

In Fig. 13, we evaluate the impact of Rician factor on the system achievable rate. It can be observed that the achievable rates of all schemes increase with the Rician factor at first and gradually saturate when the Rician factor approximates 10 dB. Moreover, the two-phase beam training scheme slightly outperforms the proposed scheme when the Rician factor is less than 5 dB. This is because our scheme is more sensitive to noise due to operations such as received-beam-pattern shifting, since NLoS components can be treated as a form of environment noise.

In Fig. 14, we plot the achievable sum-rate versus the number of users. We consider the hybrid beamforming architecture and the number of radio frequency chains is the same as the number of users. Moreover, the analog beamformer is designed to steered towards I users, while the digital beamformer is designed based on minimal mean-squared error [44]. It is observed from Fig. 14 that regardless of the number of users, the achievable sum-rate performance of the proposed near-field beam training method based on the sparse DFT codebook approaches that of the exhaustive-search method, which demonstrates the effectiveness of the proposed method extended to multi-user scenario.

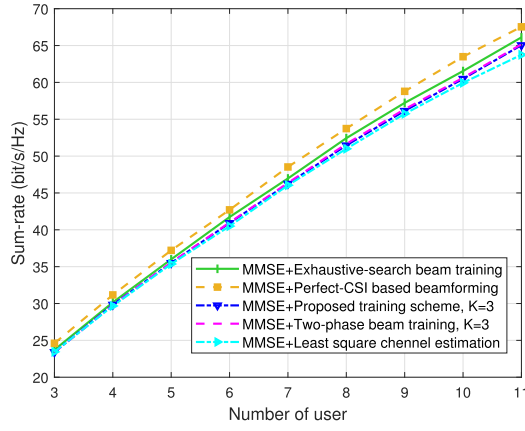


Fig. 14. Sum-rate versus number of users.

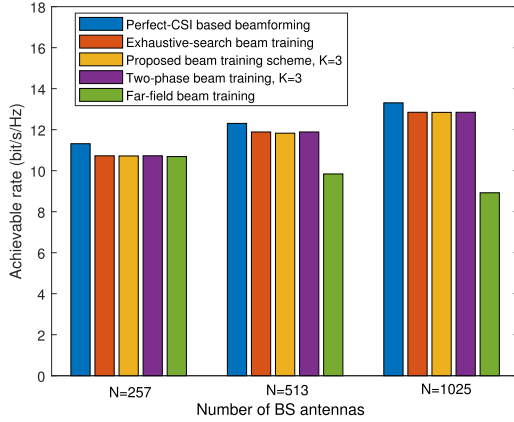


Fig. 15. Achievable rate versus number of antennas.

Finally, we plot the achievable rate versus the number of XL-array antennas in Fig. 15, where the user ranges are fixed at 80 m. It is observed that the achievable rates increase as there are more antennas for all schemes except for the far-field beam training scheme. This is because as the number of antennas increases, the near-field effect is more prominent, and hence the far-field beam training is no longer effective even when the user is located at a relatively far range (i.e., 80 m) from the XL-array.

VII. CONCLUSION

In this paper, we proposed a novel near-field beam training scheme enabled by the sparse DFT codebook (sparse far-field beamforming vectors) to construct periodic received beam pattern at the user. To this end, we showed that the angular periodicity of the received beam pattern boosts reduction in sweeping space, thereby significantly decreasing beam training overhead. Specifically, the middle of the angular support within a period contains the user angle information. Then, an activated central subarray can resolve the angular ambiguity followed by polar-domain codebook sweeping in the best user angle. Finally, numerical results were presented to show that the proposed beam training scheme can achieve nearly the same performance in the high-SNR regime with the exhaustive-search scheme, while significantly reducing the beam training overhead.

APPENDIX A

PROOF OF LEMMA 2

From (18), when $\Delta \in [-1/U, 1/U)$, B_1 is not a constant at $2k\pi$ for arbitrary integer k as q changes. Hence, we have

$$\begin{aligned} \hat{f}(r_0, \theta_0; \theta) &= \frac{1}{Q} \left| \sum_{q \in \mathcal{Q}} \exp(j\pi(A_1 q + A_2 q^2)) \right| \\ &= \frac{1}{Q} \left| \sum_{q \in \mathcal{Q}} \exp\left(j\pi A_2 \left(q + \frac{A_1}{2A_2}\right)^2\right) \right| \end{aligned} \quad (43)$$

where $A_1 = U\Delta$ and $A_2 = \frac{(Ud_0)^2}{\lambda} \frac{1-\theta_0^2}{r_0}$.

Then, the summation in (43) can be approximated by an integral, which is given by

$$\begin{aligned} \hat{f}(r_0, \theta_0; \theta) &\approx \frac{1}{Q} \left| \int_{-\frac{Q}{2}}^{\frac{Q}{2}} \exp(j\pi(A_1 q + A_2 q^2)) dq \right| \\ &\stackrel{(b_1)}{\approx} \frac{1}{Q} \left| \int_{-\frac{Q}{2}}^{\frac{Q}{2}} \exp(j\pi(A_1 q + A_2 q^2)) dq \right| \\ &\stackrel{(b_2)}{=} \frac{1}{Q} \left| \frac{1}{\sqrt{2A_2}} \int_{\sqrt{2A_2}(-\frac{Q}{2} + \frac{A_1}{2A_2})}^{\sqrt{2A_2}(\frac{Q}{2} + \frac{A_1}{2A_2})} e^{j\frac{\pi t^2}{2}} dt \right| \\ &= \left| \frac{\int_0^{\sqrt{2A_2}(\frac{Q}{2} + \frac{A_1}{2A_2})} e^{j\frac{\pi t^2}{2}} dt - \int_0^{\sqrt{2A_2}(-\frac{Q}{2} + \frac{A_1}{2A_2})} e^{j\frac{\pi t^2}{2}} dt}{\sqrt{2A_2}Q} \right|, \end{aligned} \quad (44)$$

where (b_1) is due to the approximation from the summation to the integral and (b_2) is obtained by setting $A_2(q + \frac{A_1}{2A_2})^2 = \frac{t^2}{2}$. Let $\beta_1 = \frac{A_1}{2A_2} = \Delta \sqrt{\frac{r_0}{d_0(1-\theta_0^2)}}$ and $\beta_2 = \frac{\sqrt{A_2}Q}{2} = \frac{QU}{2} \sqrt{\frac{d_0(1-\theta_0^2)}{r_0}}$, (44) can be simplified as

$$\begin{aligned} \hat{f}(r_0, \theta_0; \theta) &= \left| \frac{\int_0^{\beta_1+\beta_2} e^{j\frac{\pi t^2}{2}} dt - \int_0^{\beta_1-\beta_2} e^{j\frac{\pi t^2}{2}} dt}{2\beta_2} \right| \\ &= G(\beta_1, \beta_2), \end{aligned} \quad (45)$$

where $G(\beta_1, \beta_2) \triangleq (\hat{C}(\beta_1, \beta_2) + j\hat{S}(\beta_1, \beta_2))/(2\beta_2)$, $\hat{C}(\beta_1, \beta_2) \triangleq C(\beta_1 + \beta_2) - C(\beta_1 - \beta_2)$ and $\hat{S}(\beta_1, \beta_2) \triangleq S(\beta_1 + \beta_2) - S(\beta_1 - \beta_2)$. Specifically, $C(x) = \int_0^x \cos(\frac{\pi}{2}t^2)dt$ and $S(x) = \int_0^x \sin(\frac{\pi}{2}t^2)dt$ are the Fresnel integrals. The proof of Lemma 2.

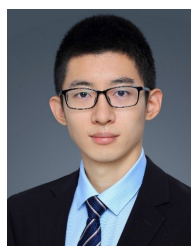
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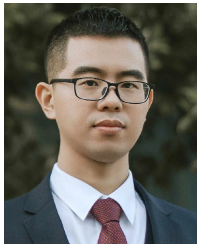
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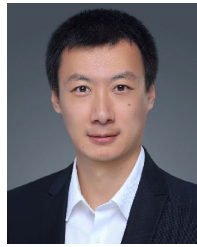


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